µInternship Report
Evaluating the pressure sensor BMP581

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Key Findings:

Pressure noise = 0.586 PaRMS

Recommended OSR = 16x

Relative Altitude Resolution = ± 0.5 m

Oxford, 26/06/2026

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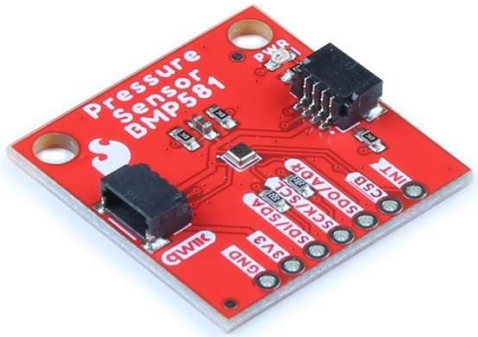
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Introduction

Bosch BMP581 is a low-power air pressure sensor designed for altitude tracking applications.
We are interested in this sensor in the context of asset tracking applications, in order to improve the **accuracy of the Z-position** of our PV trackers.

The objectives of the project are the following:

- Use BMP581 to evaluate the altitude of current position
- Explore different modes of BMP581 and calculate energy consumption required
- Configure the pressure sensor for maximum energy efficiency.
- Proof of Concept of detecting the relative position from a given reference.

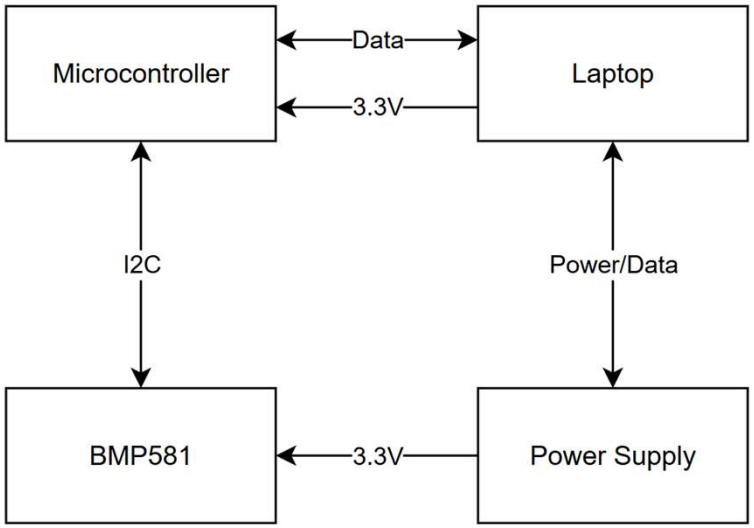


Component Selection: SparkFun Pressure Sensor BMP581

https://www.sparkfun.com/sparkfun-pressure-sensor-bmp581-qwiic.html

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Methodology



Microcontroller selection – Arduino MKR Zero

Power supply selection – Nordic PPK2

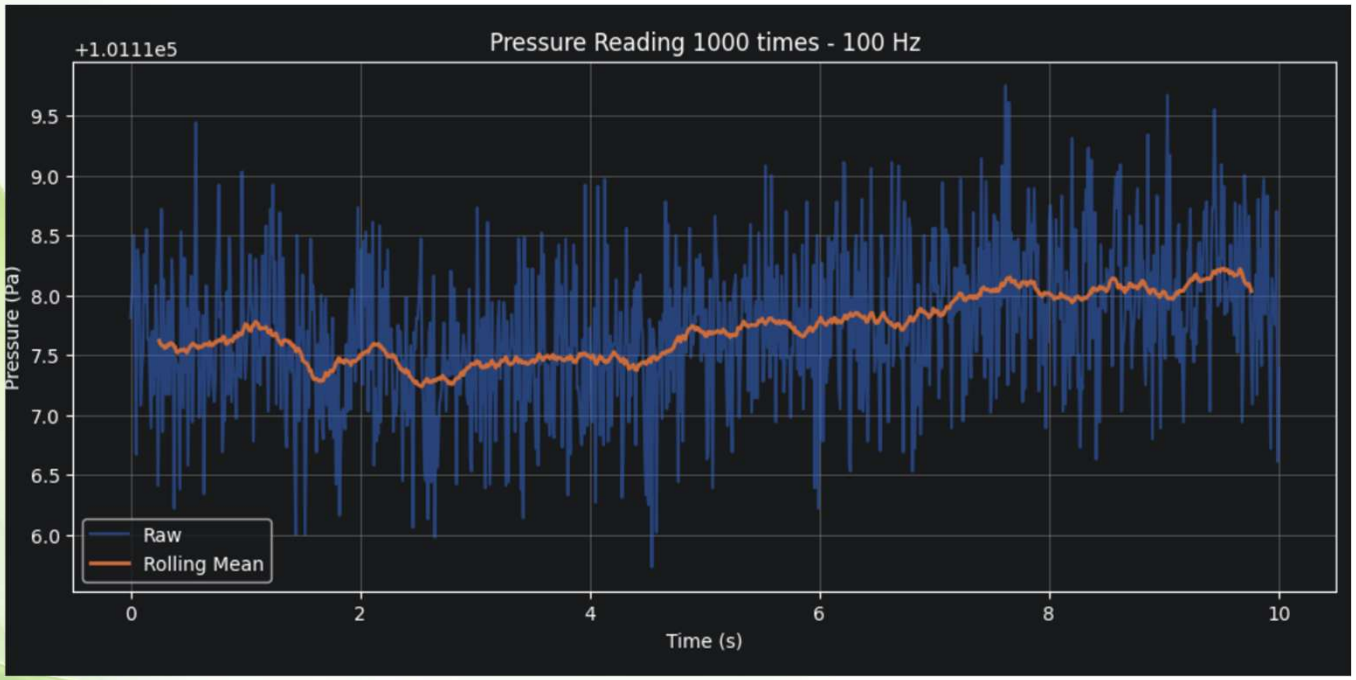
Operational Voltage – 3.3V

Microcontroller is powered by the laptop, while the BMP581 is powered separately by PPK2 in order to measure the power consumption.

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Precision Test

Precision of Sensor: A total of 1000 pressure readings at the same position, 100 readings per second



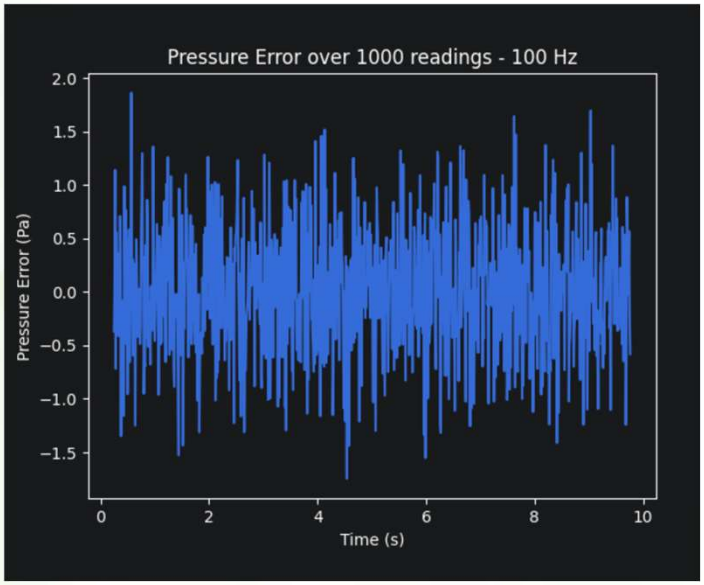
Test Condition:
- Mode: NORMAL
- ODR: 240 Hz
- Data Request Rate: 100 Hz

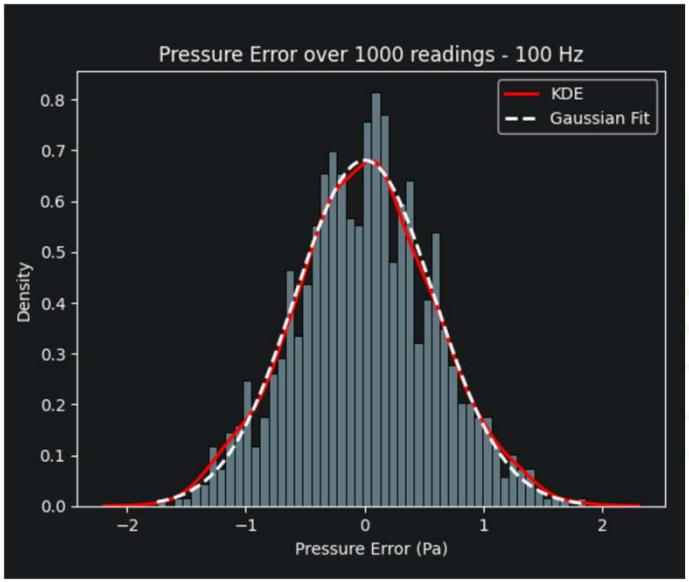
Conclusion:
- Shows a **random noise**
- A small **systematic drift** – due to non- constant true pressure for various reasons: air conditioning, door opening, etc.

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Error Distribution

Detrending the data: Error = Pressure Reading – Rolling Mean (assuming mean is the true pressure)





Mean = **0.0012 Pa**, Standard Deviation = **0.586 PaRMS**, 99% error interval = **(-1.51, 1.51) Pa**

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Comments on Precision

1. The noise is approximately **Gaussian**.

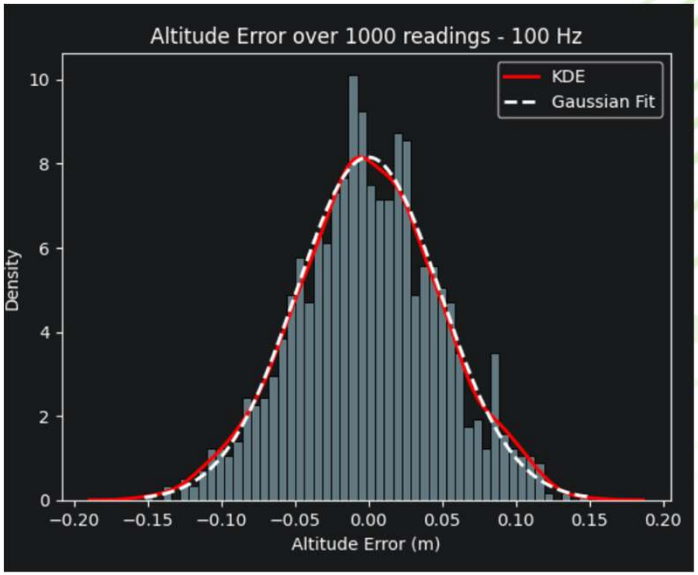
2. Pressure noise = **0.586 PaRMS**, better than typical datasheet value (**0.95 PaRMS**)

3. Altitude error distribution is also Guassian but **lower STD** (more precise). Suitable for high-resolution altitude measurements.

ISA Barometric Formula:

$$h = 44330 * [1 - (\frac{P}{P_0})^{0.19029}]$$

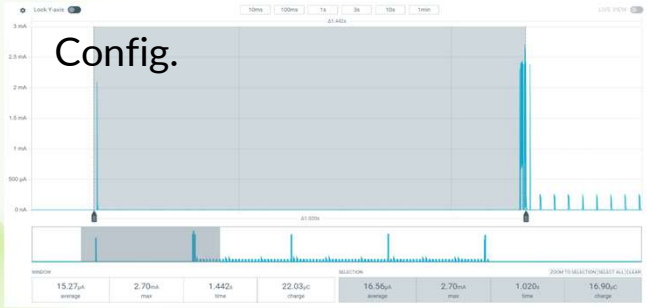
h – Altitude (m)
P – Absolute Pressure at position
P0 – Reference Pressure (Atmospheric if absolute altitude)

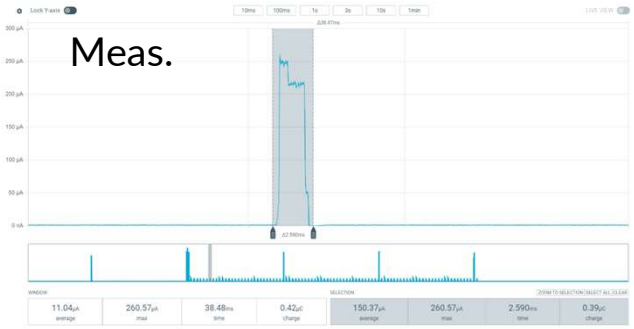


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One-reading Measurement

Test Condition:
ODR: 30Hz OSR: 1x







How much energy is required for one measurement then turn off?

Charge consumed for configuration: **16.9 uC**
Charge per Sensor Measurement: **0.4 uC**
Charge per Arduino Reading Request: **1.4 uC**

Assuming Arduino taking reading immediately after data ready (no standby)
Energy sum = Configuration + Measurement + Reading = **61.71 uJ**

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Multiple Reading Measurements

Testing standby current in basic setting, with following conditions:
ODR: 30Hz, OSR: 1x







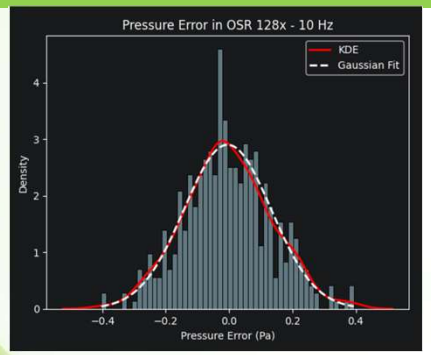
Average standby current = **1.2 uA** (typical datasheet value = **1.0 uA**)
Charge per Meas. = **0.4 uC** Charge per Reading = **1.4 uC**

In order to reach a minimum power consumption, the ODR of BMP581 should be set equal, or slightly larger than the required data reading rate.

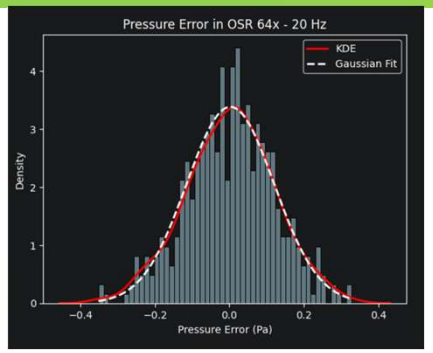
E.g. Arduino requires data at 1Hz
Minimum Power happens at ODR = 1 Hz
I = 1.2 + 0.4 + 1.4 = 3 uA, P = 3 * 3.3 = **9.9 uW + 55.77 uJ** configuration consumption

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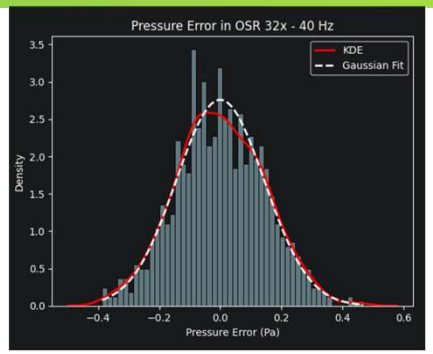
Impact of OSR on Precision



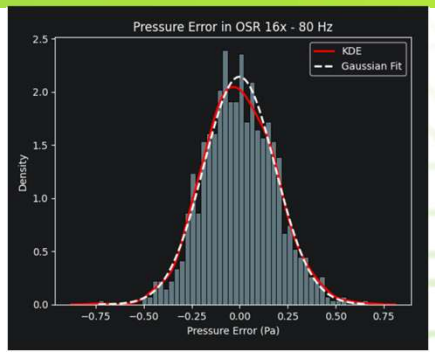
Mean = -0.0046, Std = **0.138**, 99% interval ± 0.351



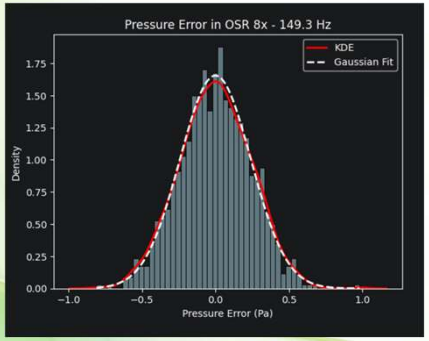
Mean = 0.0026, Std = **0.118**, 99% interval ± 0.307



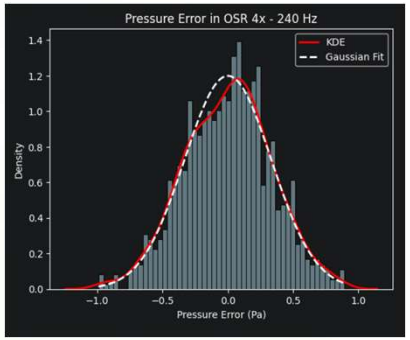
Mean = -0.0002, Std = **0.145**, 99% interval ± 0.373



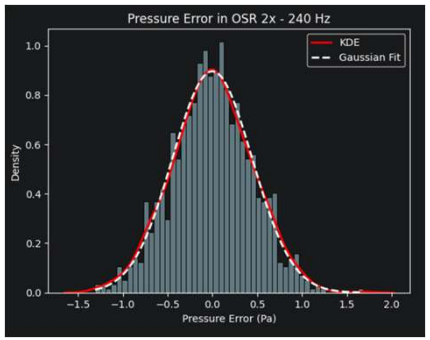
Mean = -0.0017, Std = **0.186**, 99% interval ± 0.478



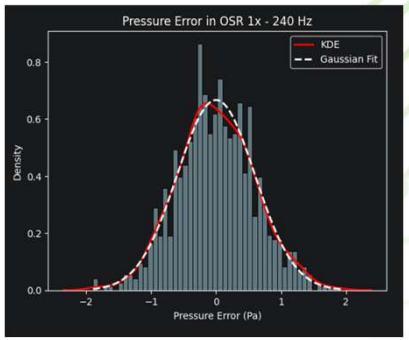
Mean = -0.0008, Std = **0.241**, 99% interval ± 0.620



Mean = -0.0010, Std = **0.333**, 99% interval ± 0.857



Mean = -0.0019, Std = **0.445**, 99% interval ± 1.14



Mean = -0.0005, Std = **0.599**, 99% interval ± 1.54

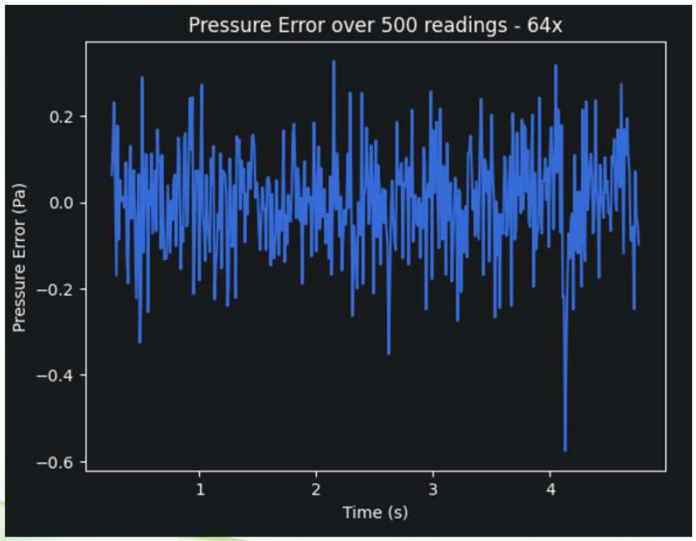
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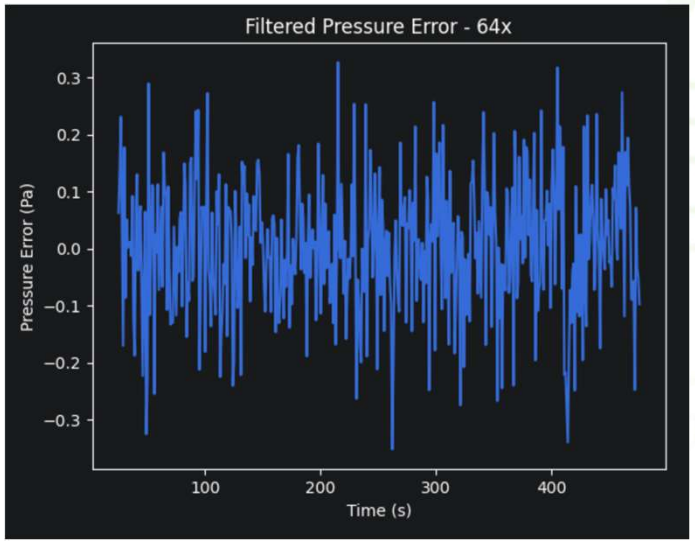
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Comments on OSR Precision

- Sensor becomes **more precise** with higher OSR, due to a significant decrease in STD.

- Abnormal data is observed particularly in OSR 128x and 64x. Filtering is applied to remove them, but further investigations should be carried out to explain the source of abnormal data in OSR 128x and 64x.





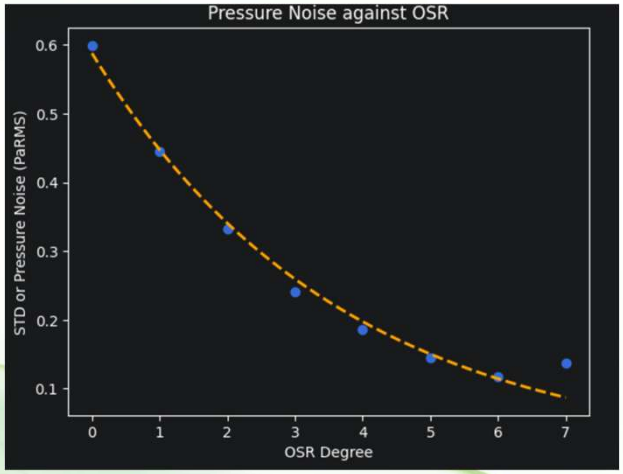
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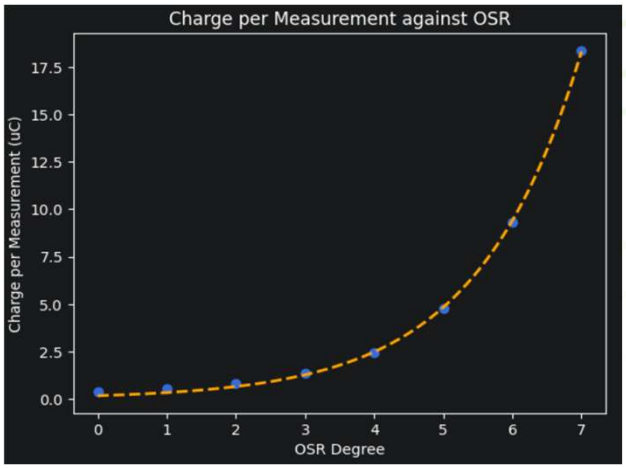
Comments on OSR Efficiency

- **Lower standby current** and (~0.3 - ~0.4 uA) and a **lower charge per Arduino reading** (0.7 uC) are observed in OSR mode. Thus, OSR with interrupt is recommended, but more studies should investigate why they are lower, especially on why standby current is smaller than typical datasheet value. (0.55 uA deep standby current)

- **Trade-off** exists between precision of result and charge consumed. **OSR 16x** is highly recommended, as most amount of pressure noise is reduced (77.7%), with only a slight (11.5%) overall increase in charge. (all compared with OSR 128x).



OSR Degree d means OSR 2^d



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Relative Altitude Measurement

Proof of Concept: Relative altitude measured between two BMP581 sensors.

Pressure at ground measured at 23/06/2026
Sensor1: 101206 Pa, Sensor2: 101222 Pa, Offset value: 16 Pa

Estimated height of first floor: **4.4 m**.

"True" First floor height: 23 stair * 0.18 m = **4.14 m**. (6.3 % error)

However, accuracy only holds for a short period of time, as the reference pressure of ground floor keeps changing, strongly affecting the result.

Quick tests also show that BMP581 can support up to **± 0.5 m** of altitude change.

Altitude output from Arduino

```
Time:
47
Pressure (Pa): 101153.50
Reference Pressure Used (Pa): 101222.00
Offset Pressure (Pa): 16.00
Altitude (m): 4.38


Time:
48
Pressure (Pa): 101153.22
Reference Pressure Used (Pa): 101222.00
Offset Pressure (Pa): 16.00
Altitude (m): 4.40
```


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Limitations and Future Work

1. The method used to test the BMP581 sensors could be improved. A **sealed chamber** with maintained pressure is highly recommended.

2. More investigations need to be carried out with **OSR configuration**, especially in its lower standby current and abnormality of 128x and 64x ratio.

3. It is proved that relative altitude can be accurately measured, but in a short period of time. Further improvements can be implemented, such as using two separate Arduinos and constantly exchaning data via **wireless communication**. Thus, valid reference pressure can be always available.

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